

Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 2: Flower Detectives: Classifying Flowers by Ovary Position, Symmetry, Inflorescence and Sex

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	D	A	D	B	C	A

USE AFTER: OBSERVE PHASE — Observe: slicing the flower lengthwise to see whether the ovary sits above (superior) or below (inferior) the point where petals attach

Q1. A learner cuts a Hibiscus flower lengthwise and sees that the ovary sits above the point where the petals attach. How is this ovary described?

- A. Zygomorphic ovary
- B. Unisexual ovary
- C. Superior ovary**
- D. Inferior ovary

Correct answer: C. An ovary above the point where the petals attach is superior. An inferior ovary sits below that point, while zygomorphic describes symmetry and unisexual describes whether both male and female parts are present.

USE AFTER: OBSERVE PHASE — Observe: counting petals and sepals and deciding symmetry by cutting the flower along different lines

Q2. A learner cuts a flower in two along several different lines and gets two equal halves every time. What type of symmetry does the flower have?

- A. Unisexual, because the halves are equal
- B. Inferior, because the halves are the same size
- C. Zygomorphic, because it can be cut in more than one way
- D. Actinomorphic, because more than one line gives equal halves**

Correct answer: D. A flower is actinomorphic (regular) if more than one line gives equal halves. A zygomorphic flower gives equal halves along only one line.

USE AFTER: OBSERVE PHASE — Observe: Bougainvillea station (which parts are the real flowers and which are the colourful bracts)

Q3. A Bougainvillea branch has small flowers surrounded by showy, colourful parts that look like petals. What are the showy, colourful parts?

- A. Bracts, and the real flowers are the small ones inside them**
- B. Stamens, which are brightly coloured to attract insects
- C. Ovaries, which grow outside the flower
- D. Petals, and the small flowers are extra sepals

Correct answer: A. In Bougainvillea the colourful parts are bracts that surround the small true flowers. This is why the 'petals' seen at first are misleading.

USE AFTER: OBSERVE PHASE — Observe: maize inflorescence station (male tassel and female cob silks on the same plant)

Q4. A maize plant has a tassel at the top and a cob with silks lower down. Which statement about the flowers of maize is correct?

- A. The tassel carries female flowers and the silks belong to male flowers
- B. Every maize flower is bisexual, with stamens and carpels together
- C. Male and female flowers are on different maize plants
- D. The tassel carries male flowers and the silks belong to female flowers, both on the same plant**

Correct answer: D. Maize has separate unisexual flowers on one plant: the tassel has the male parts (stamens) and the silks are part of the female flowers. This is called monoecious.

USE AFTER: EXPLAIN PHASE — Explain Part B: CER on classification and the phenomenon (why a lone pawpaw or avocado tree may have many flowers but no fruit)

Q5. A farmer in Kisumu has one pawpaw tree that is covered in flowers every season but never gives fruit. Which explanation is best supported by what we know about flowers?

- A. Fruit forms from petals, and the tree has too few petals
- B. Fruit develops from a fertilised ovary, so if the flowers on the tree lack stamens or carpels, pollen or ovules must come from another tree**
- C. Only flowers with an inferior ovary can make fruit
- D. Many flowers always means many fruits, so the tree must be sick

Correct answer: B. Fruit forms from a fertilised ovary, which needs pollen from stamens. If a tree has only flowers of one sex, it needs another tree to supply the missing part. The idea that many flowers mean many fruits is a misconception.

USE AFTER: EXPLAIN PHASE — Explain Part A: Think-Pair-Share on placentation (counting chambers and seeing how ovules are attached)

Q6. A learner cuts across the ovary of a flower and sees small chambers with tiny round structures attached inside. What is the way these ovules are attached inside the ovary called?

- A. Symmetry
- B. Inflorescence
- C. Placentation**
- D. Pollination

Correct answer: C. Placentation is how ovules are attached inside the ovary. Inflorescence is how flowers are arranged on the plant, and pollination is the transfer of pollen.

USE AFTER: MODEL BUILDING PHASE — Model: adding a 'Flower' section with an arrow system showing that fruit needs pollen to reach the ovary

Q7. A group adds an arrow to its model of a flower to show how pollen must move before fruit can form. Which arrow is correct?

- A. From the anther of the stamen to the stigma of the carpel**
- B. From the stigma to the anther
- C. From the ovary to the filament
- D. From the petal to the ovary

Correct answer: A. Pollen is made in the anthers (male part) and must reach the stigma (female part) so that the ovary can be fertilised and develop into fruit.